

Stealth Fighter

Stealth is the ability to evade detection, and stealth fighters attempt to minimise their heat and radar signatures to do so.

We analyse the F-22 Raptor's – a superior stealth plane – methods of avoiding radar and heat detection.

Hoodwinking Radar

A major component of a stealth fighter's ability to go undetected over a radar field is its unconventional design. The sharp nose, angular edges and slim design has less to do with aerodynamical integrity as compared to stealth. Even the materials used to make a stealth fighter like the F-22 is different from your commercial airliner – the Boeing 747.

Here we see that the commercial 747 jumbo jet's footprint is much larger compared to the F-22. And its exterior body is unlike the F-22's, even the components and material used is different – one that is easily detected off radar.

So as radio waves are emitted from a radar dish, the ones hitting the jumbo jet reflect back to the receiving dish, whereas the same waves hit the F-22 and are either absorbed or dispersed after encountering its exterior.